Final Project 1: Genetic Algorithm for Robot Swarms (100 pts)

Background: Central-Place Foraging Algorithm (CPFA)

The CPFA algorithm is inspired by the foraging behavior of ant colonies [1]. Its main purpose is to program a swarm of robots to autonomously forage for resources (e.g., minerals or survivors) in a large unknown environment and then deliver them to the central collection zone. Initially, a robot departs from the central collection zone and searches for resources randomly. If it finds resources, it will create a virtual pheromone waypoint based on the density of resources around that location. It will share this waypoint with the central server in the collection zone when it returns to the center. Once it arrives at the center, it drops off the collected resource and can access other pheromone information in the central server. It can decide where it should go based on the shared pheromone information in the central server, the last location where it found resources (called site fidelity), or search for resources randomly again. The behavior of a single robot is shown in Figure 1.

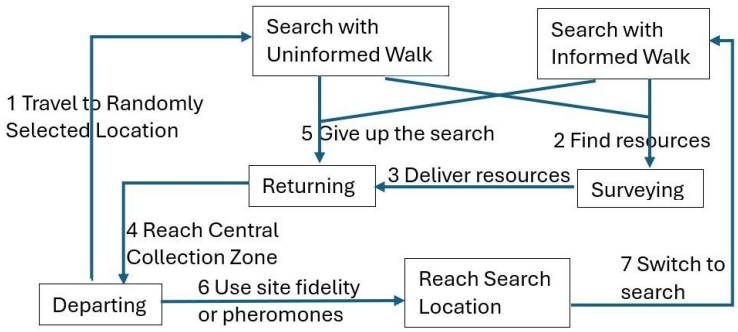


Figure 1: Individual robot's behavior in the CPFA

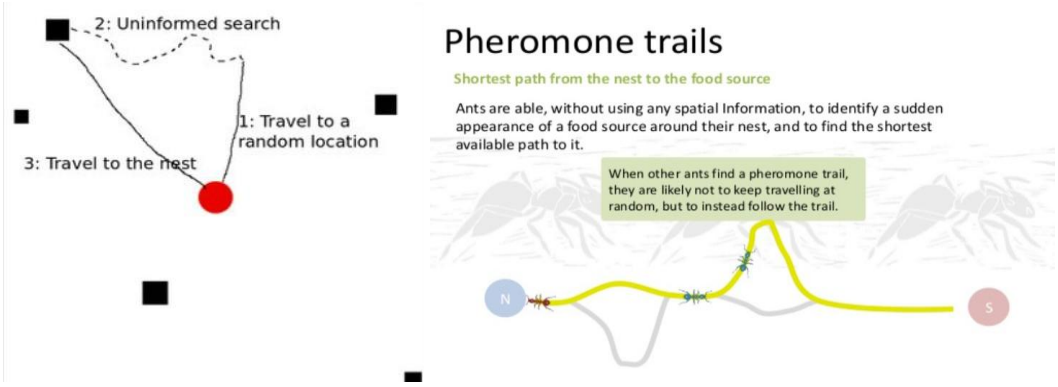


Figure 2: Sharing pheromone trails

Description:

I share my CPFA algorithm GitHub repository with you.
https://github.com/MARSLab-UTRGV/Collision_Free_CPFA/tree/main

It includes the Genetic Algorithm (GA) that can represent and optimize the 7 CPFA parameters as seen below [1]. The decision-making of robots is controlled by a set of parameters. The initial values of the parameters are generated by the set of initialization functions. The GA algorithm is implemented in Python. You can run the ga.py to train the parameters. The basic GA operations and functions are in argos_util.py.

Parameter	Description	Initialization function
p_s	Probability of switching to searching	$\mathcal{U}(0, 1)$
p_r	Probability of returning to nest	$\mathcal{U}(0, 1)$
ω	Uninformed search variation	$\mathcal{U}(0, 4\pi)$
λ_{id}	Rate of informed search decay	exp(5)
λ_{sf}	Rate of site fidelity	$\mathcal{U}(0, 20)$
λ_{lp}	Rate of laying pheromone	$\mathcal{U}(0, 20)$
λ_{pd}	Rate of pheromone decay	exp(10)

Figure 3: The set of parameters

The GA should consider the number of resources successfully deposited within some time constraint as a fitness score. One of the trained/evolved parameters is shown in the file evolved_parameters.txt in the shared GitHub repository.

Goals:

The goal is to train the 7 parameters using the GA algorithm. You need to get the best 7 parameters for each resource distribution (Random, Powerlaw, and Clustered). The following table shows the settings in the training. You will also evaluate your trained parameters in the same setting.

Table 1: Training and experimental configuration

Size of the arena (m)	10 * 10
The radius of the collection zone (m)	0.15
Number of resources	256
Resource distributions	Random Powerlaw Clustered
Number of robots	24
Foraging time (minutes)	12
Maximum generation	150

1. Let the training work and save the training data. Please show me your first training result and plot the average and standard deviation in each generation.
2. Please train 10 times for each resource distribution and select the best set of

parameters. Then, you can plot the results of the three distributions in a box plot. The data for each box plot is from 50 runs.

3. Please plot the number of resources collected each minute in a bar plot for all distributions.

I will compare your foraging scores to those of other teams. The team that has a higher score will get a higher grade.

Videos:

Multiple Place Foraging Algorithm:
https://www.youtube.com/watch?v=CoRMhYpC9gw

References

[1] Joshua Hecker and Melanie Moses. Beyond Pheromones: Evolving Error-Tolerant, Flexible, and Scalable Ant-Inspired Robot Swarms. Swarm Intelligence. March 2015. https://link.springer.com/article/10.1007/s11721-015-0104-z

[2] Melanie Moses. Toward Infinitely Scalable Robotic Swarms. Santa Fe Institute. chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://wiki.santafe.edu/images/7/73/Moses_RobotSwarms_SFI_CSSS_June_2017.pdf

[3] J. F. Frenzel. Genetic Algorithms. IEEE Potentials. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=282292

Resources:

- KieCodes: Genetic Algorithms Explained By Example

https://www.youtube.com/watch?v=uQj5UNhCPuo&list=PLuZkwckxno0qjTQcrfaQ3INlCkM4bn5fk